

B.Tech. CSE (III YEAR - V SEM) (2025-2026)
**DEPARTMENT OF COMPUTER ENGINEERING &
APPLICATIONS**



**AI-Based Emotion Recognition & Mental
Health Monitoring**

Team ID: T7

Team Members

1. **Priyam Kumar**– *University Roll No: 2315510152*
2. **Piyush Maurya**– *University Roll No: 2315510143*
3. **Nakul Upadhyay**– *University Roll No: 2315510130*

Mentor : Ms. Anushka Shukla

Signature: _____

Project Description (Introduction):

The **AI-Based Emotion Recognition & Mental Health Monitoring system** is an **innovative application** designed to detect and analyze **human emotions in real time** using **Artificial Intelligence (AI)**, **Computer Vision**, and **Natural Language Processing (NLP)**. The project aims to assist **therapists, educators, and organizations** by providing deeper insights into user emotions through **facial expressions** and **voice analysis**.

In today's fast-paced digital world, **mental health challenges** are increasing, yet **real-time emotion tracking** during **online therapy or interactions** remains limited. This system bridges that gap by combining **facial emotion recognition** and **speech sentiment analysis** to accurately identify emotions such as **happiness, sadness, anger, surprise, and neutrality**. The system continuously monitors **emotional trends** and displays **analytics** through a **secure cloud-based dashboard**.

The proposed solution uses advanced **AI models** — such as **Convolutional Neural Networks (CNN)** for **facial emotion detection** and **1D Convolutional Neural Networks (Conv1D)** for **speech emotion analysis**. The outputs from both modalities are **fused** to produce a more **accurate emotional state**, ensuring better reliability than single-mode systems.

Users can interact with the system through a **web dashboard** built using modern web technologies like **React, Node.js**, and **cloud services (AWS/GCP/Azure)**. The dashboard visualizes **emotion trends**, generates **daily and weekly reports**, and even triggers **alerts** in case of **stress or emotional anomalies**. This feature is especially beneficial for **therapists monitoring patient progress remotely** or **educators assessing student engagement**.

The project emphasizes **data security** and **compliance** with global standards such as **GDPR** and **HIPAA**, ensuring that all **video and audio data** are **encrypted** and **stored safely**. It provides a **non-invasive, ethical, and privacy-focused** approach to **emotional analytics**.

By integrating **AI-driven emotion detection** with **mental health monitoring**, the system empowers professionals to make more **informed decisions**, promotes **emotional well-being**, and enhances **digital human interaction**. Overall, the project represents a step toward **emotionally intelligent computing** that supports **mental health awareness** and **proactive care**.

Table of Content

S.No.	Content	Page No.
1	Introduction	1
2	Index	2
3	Detailed Week-wise Timeline	3
4	Individual Contributions	4–5
5	Technical Documentation & Research	6–7
6	Snapshots (Code, Output, UI)	8–9

Detailed Week-Wise Timeline (with Individual Tasks)

Week	Dates	Milestones	Priyam Kumar (Lead)	Nakul Upadhyay (Frontend)	Piyush Maurya (Backend)	Deliverables	Status
1	1 Sep – 15 Sep	Requirement gathering, problem definition	Finalize scope	Explore cloud DB & hosting services (AWS/GCP)	Draft SRS, Project scope doc	Draft SRS, Project scope doc	Completed
2	16 Sep – 22 Sep	Literature survey & architecture design	Architecture review	Database schema, API design	Research report, workflow diagrams	Research report, workflow diagrams	Completed
3	23 Sep – 5 Oct	Facial emotion recognition module	Monitor risks, progress tracking	Backend service for image/video input storage	Facial model prototype	Facial model prototype	Completed
4	6 Oct – 12 Oct	Speech sentiment analysis module	Ensure module alignment, review progress	API integration for audio processing	Speech model prototype	Speech model prototype	Completed
5	13 Oct – 28 Oct	Multi-modal fusion model	Document fusion approach	Integrate APIs for model outputs	Fusion results & accuracy test	Fusion results & accuracy test	In Progress
6	29 Oct – 15 Nov	Cloud dashboard	Define KPIs, usability checks	Connect dashboard to backend + cloud DB	Interactive dashboard	Interactive dashboard	Completed
7	15–22 Nov	Testing & refinement	Review test cases, compile reports	Unit testing (AI models, UI)	Integration testing (backend, APIs, storage)	Test report	In Progress

Week	Dates	Milestones	Priyam Kumar (Lead)	Nakul Upadhyay (Frontend)	Piyush Maurya (Backend)	Deliverables	Status
8	22–29 Nov	Final deployment	Final sign-off	Dashboard polish, model fine-tuning	Deployment on cloud, prepare release notes	v1.0 Release + Presentation	In Progress
9	29 Nov – 6 Dec	Post-deployment analysis & feedback	Collect feedback, document report	UI/UX improvements	Backend monitoring & bug fixing	Final report submission	Pending

Individual Contributions

Priyam Kumar (Team Lead / Machine Learning Engineer)

- Developed and trained a **Facial Emotion Recognition model** using **TensorFlow (Keras)** and **Computer Vision techniques**, leveraging **CNN-based architectures** on the **FER-2013 dataset** for real-time emotion classification.
- Built and optimized a **Speech Emotion Recognition model** using **1D Convolutional Neural Networks (Conv1D)** on the **RAVDESS dataset**, focusing on MFCC feature extraction and temporal pattern learning.
- Preprocessed and balanced emotion datasets (**FER-2013** for facial and **RAVDESS** for speech) by applying augmentation, normalization, and encoding techniques to ensure consistent and high-quality training input.
- Managed **project coordination, architecture design**, and ensured alignment of all modules with the defined timeline.

GitHub Branch: Documentation-&-Model

Nakul Upadhyay (Frontend Engineer)

- Built an interactive **React-based cloud dashboard** for real-time emotion visualization, reports, and analytics.
- Designed the **user interface (UI/UX)** for teletherapy integration, ensuring smooth camera and microphone access.
- Implemented **data visualization components** (charts, emotion trend graphs) using APIs connected to backend services.

- Contributed to **frontend testing, debugging**, and ensured responsive, accessible web design.

GitHub Branch: frontend

Piyush Maurya (Backend / Cloud & Integration Engineer)

- Designed and implemented **secure backend services** using **Node.js, Express, and MongoDB** for data management.
- Handled **API integrations** between emotion recognition models and the dashboard, ensuring seamless communication.
- Implemented **authentication, session management**, and ensured compliance with **HIPAA** principles for user data.
- Managed **integration testing, cloud monitoring**, and prepared the final **deployment and release documentation**.

GitHub Branch: backend

Technical Documentation & Research

Technologies Used

Component	Technology	Purpose
Frontend	React, HTML, CSS, JavaScript	UI for document upload & result display
Backend	Flask (Python)	Handles API requests, emotion detection logic, and model integration
AI Models	CNN	CNN for facial emotion recognition, 1D CNN for speech emotion recognition
Database	MongoDb	Stores user profiles, emotion logs, and session history
Deployment	Docker	Containerization & portability

Research & Concepts

- Used **Convolutional Neural Networks (CNNs)** to extract spatial features from facial expressions in images and videos, enabling accurate detection

and classification of emotions such as happiness, sadness, anger, and surprise.

- Developed **Flask-based REST APIs** to facilitate seamless communication between the frontend and backend systems.
- Applied **model optimization techniques**, including hyperparameter fine-tuning and overfitting reduction, to enhance performance and reliability.
- Implemented **Docker** for containerization, ensuring smooth deployment and consistent cross-platform execution.

Snapshots

Figure1:Personalized Care Mode Selection – Welcome screen where users choose between Wellness, Support, or Clinical modes, each designed with

smart AI tools for personalized mental health support.

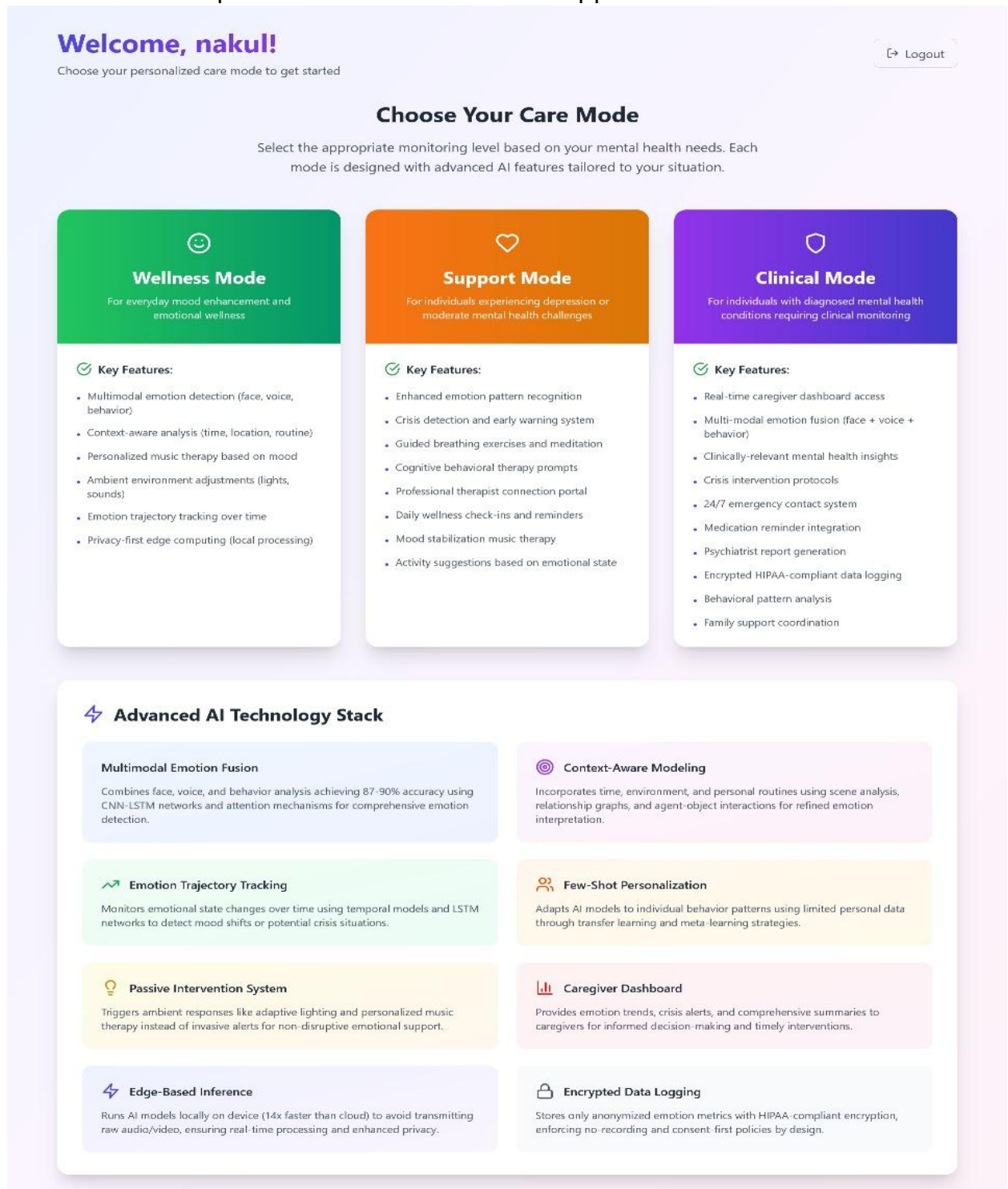


Figure2:Multimodal Emotion Scan Interface – Interactive screen allowing users to start emotion scans through face, voice, and behavior analysis while ensuring complete privacy and local data processing.

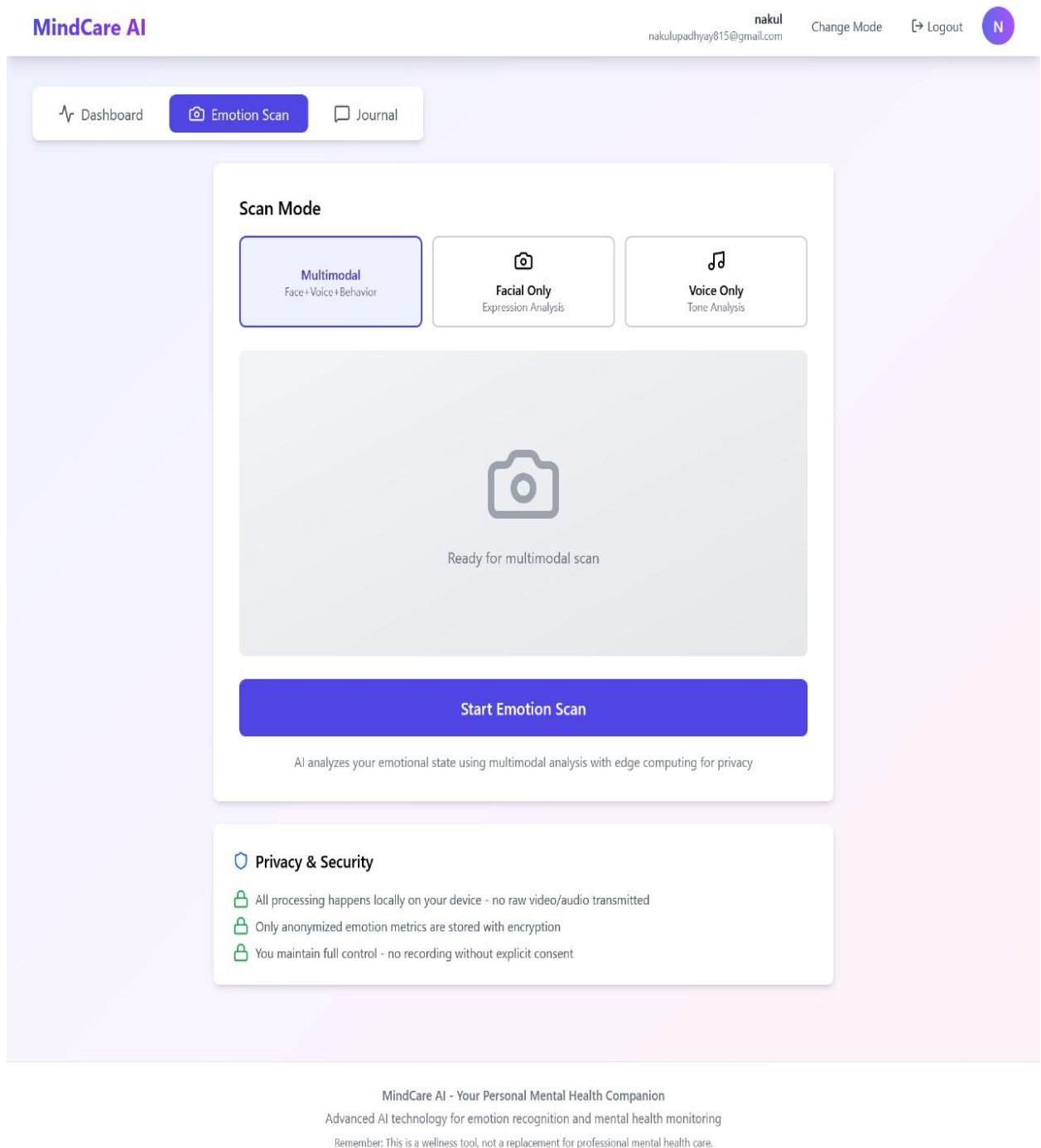


Figure3:Support Mode Dashboard Overview – Insightful dashboard showing the user’s emotional wellness score, AI accuracy, and available support activities like breathing exercises and CBT prompts.

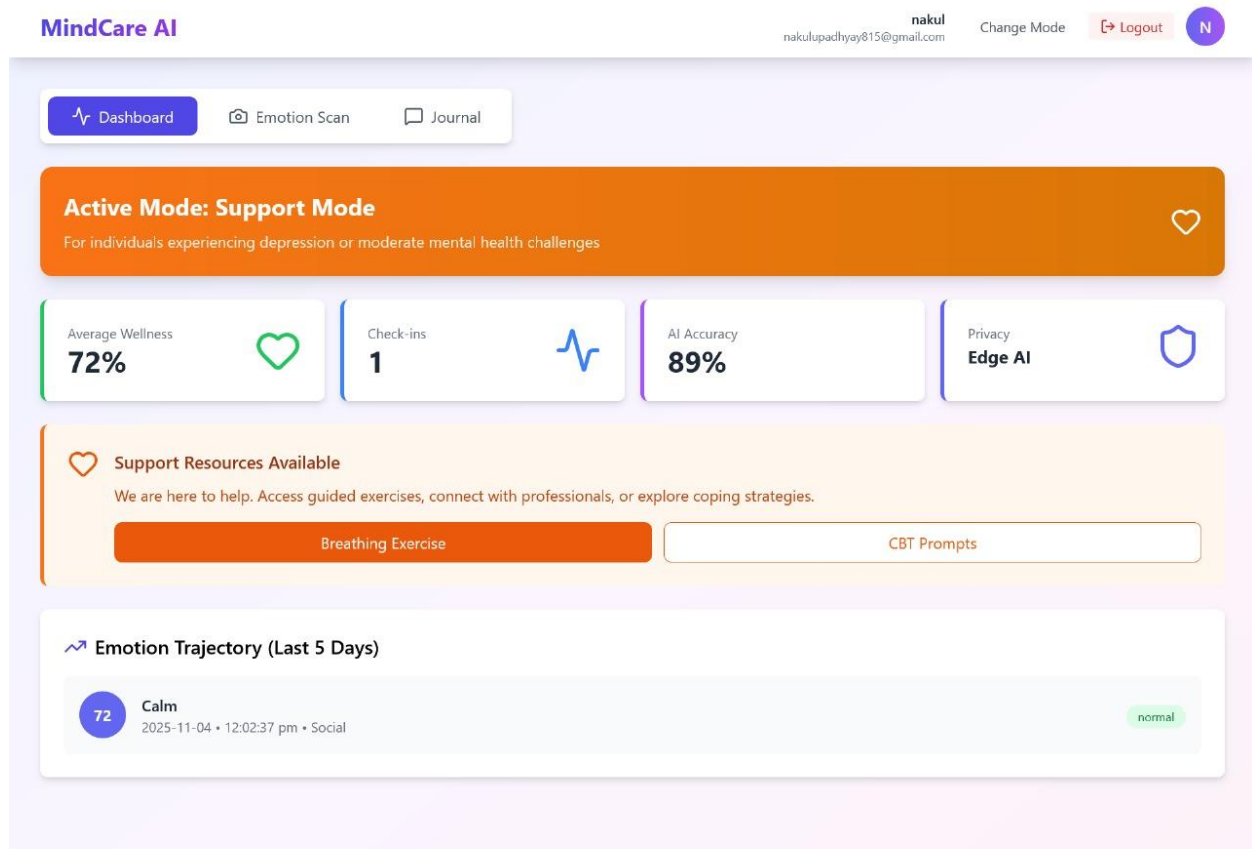


Figure4:Mood Journal & History Tracker – Simple and clean view summarizing past mood entries, emotion trends, and context-based insights for self-reflection

and progress tracking.

MindCare AI

nakul
nakulupadhyay815@gmail.com

Change Mode

Logout

N

Dashboard

Emotion Scan

Journal

Mood Journal & History

Calm

2025-11-04 • 12:02:37 pm

normal

72%

Quick emotion check

Context: Social

MindCare AI - Your Personal Mental Health Companion

Advanced AI technology for emotion recognition and mental health monitoring

Remember: This is a wellness tool, not a replacement for professional mental health care.